Name: _____ **Unit 2:** Equations & Inequalities

Date: _____ **Bell:** _____ **Homework 4:** Proportions

Directions: Solve each equation. **SHOW ALL STEPS!**

1. $\frac{x}{4}=\frac{3}{2}$

$2x = 4\cdot 3$

$\frac{2x}{2} = \frac{12}{2}$

$\boxed{x=6}$

2. $\frac{f+3}{12}=\frac{7}{2}$

$2(f+3) = 12\cdot 7$

$2f+6 = 84$

$-6 \quad -6$

$\frac{2f}{2} = \frac{78}{2}$

$\boxed{f=39}$

3. $\frac{4}{n+9}=\frac{2}{7}$

$2(n+9) = 4\cdot 7$

$2n+18 = 28$

$-18 \quad -18$

$\frac{2n}{2} = \frac{10}{2}$

$\boxed{n=5}$

4. $\frac{3}{8}=\frac{s-2}{24}$

$8(s-2) = 3\cdot 24$

$8s-16 = 72$

$+16 \quad +16$

$\frac{8s}{8} = \frac{88}{8}$

$\boxed{s=11}$

5. $\frac{3}{10}=\frac{b+7}{20}$

$10(b+7) = 3\cdot 20$

$10b+70 = 60$

$-70 \quad -70$

$\frac{10b}{10} = \frac{-10}{10}$

$\boxed{b=-1}$

6. $\frac{3}{16y}=\frac{1}{y+13}$

$16y = 3(y+13)$

$16y = 3y+39$

$-3y \quad -3y$

$\frac{13y}{13} = \frac{39}{13}$

$\boxed{y=3}$

7. $\frac{m}{3}=\frac{m+4}{7}$

$7m = 3(m+4)$

$7m = 3m+12$

$-3m \quad -3m$

$\frac{4m}{4} = \frac{12}{4}$

$\boxed{m=3}$

8. $\frac{a}{2}=\frac{a-4}{6}$

$2(a-4) = 6a$

$2a-8 = 6a$

$-2a \quad -2a$

$\frac{-8}{4} = \frac{4a}{4}$

$\boxed{a=-2}$

9. $\frac{x+2}{3}=\frac{x+4}{5}$

$5(x+2) = 3(x+4)$

$5x+10 = 3x+12$

$-3x \quad -3x$

$2x+10 = 12$

$-10 \quad -10$

$\frac{2x}{2} = \frac{2}{2}$

$\boxed{x=1}$

10. $\frac{n+8}{5}=\frac{n-1}{2}$

$2(n+8) = 5(n-1)$

$2n+16 = 5n-5$

$-2n \quad -2n$

$16 = 3n-5$

$+5 \quad +5$

$\frac{21}{3} = \frac{30}{3}$

$\boxed{n=7}$

11. $\frac{p-5}{p+1}=\frac{2}{3}$

$3(p-5) = 2(p+1)$

$3p-15 = 2p+2$

$-2p \quad -2p$

$p-15 = 2$

$+15 \quad +15$

$\boxed{p=17}$

12. $\frac{x+2}{6}=\frac{2x-10}{8}$

$8(x+2) = 6(2x-10)$

$8x+16 = 12x-60$

$-8x \quad -8x$

$16 = 4x-60$

$+60 \quad +60$

$76 = 4x$

$\boxed{x=19}$

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